

Sulfa drugs are any of a class of synthetic chemical substances derived from sulfanilamide or para-amino benzene sulfonamide. They are used to treat bacterial infections, although they have largely been replaced for this purpose by antibiotics. Some are also used in the treatment of diabetes. Some of the sulfa drugs are Sulfonamide antibiotics sulfadiazine, sulfasalazine, sulfoxazole, sulfacetamide etc. Prontosil was the first sulfa drug discovered in 1932.

42. Sulpha drug is which of the following types of drug?

- (a) Analgesics (b) Antibacterial
(c) Antiseptic (d) Antipyretic

U.P. P.C.S. (Pre) 2022

Ans. (b)

See the explanation of above question.

43. Which one of the following is a Sulpha Drug?

- (a) Penicillin (b) Streptomycin
(c) Cibazole (d) Aspirin

U.P. P.C.S. (Mains) 2017

Ans. (c)

Cibazole is a short-acting sulpha drug. It was used to be as a common oral and topical antimicrobial until less toxic alternatives were discovered. It is still occasionally used, sometimes in combination with sulfabenzamide and sulfacetamide. Cibazole is effective against a wide range of gram-positive and gram-negative pathogenic micro-organisms. Although no longer used in humans, it is used in cattle. Penicillin and streptomycin are antibiotics, while Aspirin is antipyretic medicine.

44. Which of the following are the reasons for the occurrence of multi-drug resistance in microbial pathogens in India?

1. Genetic predisposition of some people
2. Taking incorrect doses of antibiotics to cure diseases
3. Using antibiotics in livestock farming
4. Multiple chronic diseases in some people

Select the correct answer using the code given below:

- (a) 1 and 2 only (b) 2 and 3 only
(c) 1, 3 and 4 (d) 2, 3 and 4

I.A.S. (Pre) 2019

Ans. (b)

According to 'Scoping Report on Antimicrobial Resistance in India' released by the Department of Biotechnology (DBT) in November 2017, main reasons for the occurrence of multi-drug resistance (MDR) in microbial pathogens in India are :

- (i) High consumption of broad-spectrum antibiotics;
- (ii) Injudicious use of antibiotic fixed-dose combinations;
- (iii) Self-medication, access to antibiotics without prescription and taking incorrect doses of antibiotics to cure diseases;

- (iv) Cultural activities like mass bathing in rivers;
- (v) Heavy use of antibiotics in livestock farming;
- (vi) Pharmaceutical industry pollution;
- (vii) Poor sanitation facilities and disposal of untreated sewage in water bodies;
- (viii) Poor infection prevention practices in healthcare settings; etc.

Thus reasons given in statements 2 and 3 are correct.

III. Fungal Diseases

Notes

- The Fungi are a separate kingdom of living things.
- A fungus (plural fungi) is a kind of living organisms; yeast, moulds (molds) and mushrooms are types of fungi.
- Fungi have cells with nuclei (eukaryotic organisms).
- Fungus cell wall contains chitin unlike the cell wall of plants, which contain cellulose.
- Most fungi grow as tubular filaments called **hyphae**. An interwoven mass of hyphae is called a **mycelium**. The walls of hyphae are often strengthened with chitin, a polymer of **N-acetylglucosamine**.
- The study of fungi is called **Mycology**.
- All fungi are heterotrophs either **parasitic** or **saprotrophs**.
- Fungi have a close relation to humans. Several species of fungi are useful and harmful to humans.
- Some main fungal infections have been described as below :

Aflatoxins :

- Aflatoxins are a family of toxins produced by certain fungi that are found on agricultural crops such as maize, peanuts, cotton seeds and tree nuts.
- The main fungi that produce aflatoxins are **Aspergillus flavus** and **Aspergillus parasiticus**, which are abundant in warm and humid regions of the world.
- Aflatoxin-producing fungi can contaminate crops in the field, at harvest, and during storage.
- Aflatoxin is a natural **carcinogen** which is associated with an increased risk of **liver cancer**.
- Aflatoxin causes DNA damage and with prolonged exposure to aflatoxin, cells accumulate DNA mutations and thus are at increased risk of developing into cancer cells.

Aspergillosis :

- Aspergillosis is the name given to a wide variety of diseases caused by infection by **fungi** of the genus **Aspergillus**.

- Aspergillosis occurs in humans, birds and other animals.
- The majority of the cases occur in people with an underlying illness such as **tuberculosis** or chronic **obstructive pulmonary disease**.

Ergotism :

- Ergotism is the effect of long-term ergot poisoning, traditionally due to the ingestion of the **alkaloids** produced by the **Claviceps purpurea** fungus that infects rye and other cereals and more recently by the action of a number of **ergoline** based drugs.
- It is also known as **ergototoxicosis**, **ergot poisoning** and **Saint Anthony's Fire**.
- Toxins affect the nervous, vascular (vasoconstriction and resultant gangrene), and endocrine system.
- Clinical signs include decreased food intake and growth, necrosis of beak, comb and toes, distorted feather development and enteritis.
- Ergotism primarily affects leghorns and other laying breeds and only trace amounts remain in their meat.

Athlete's Foot :

- Athlete's foot (**Tinea pedis**) is a fungal infection that usually begins between the toes. It affects the upper layer of the skin of the foot.
- It commonly occurs in people whose feet have become very sweaty while confined with tightfitting shoes.
- Signs and symptoms of athlete's foot include a scaly rash that usually causes itching, cracking, stinging and burning.
- Athlete's foot is contagious and can be spread via contaminated floors, towels or clothing.
- The fungal infection is called athlete's foot because it is commonly seen in athletes.
- Athlete's foot is caused by the fungus Trichophyton, Epidermophyton and Microsporum.

Mucormycosis :

- **Mucormycosis** (previously called zygomycosis) is a serious but rare fungal infection caused by a group of molds called mucormycetes. These molds live throughout the environment.
- **Mucormycosis** mainly affects people who have health problems or take medicines that lower the body's ability to fight germs and sickness.
- It is also known as **black fungus**.
- Symptoms depend on where in the body the infection occurs. It most commonly infects the nose, sinuses, eye, and brain resulting in a runny nose, one-sided facial swelling and pain, headache, fever, blurred vision, bulging or displacement of the eye (proptosis), and tissue death.
- Other forms of disease may infect the lungs, stomach and intestines, and skin.

Dermatitis, baldness, candidiasis, geotrichosis are the fungal borne diseases.

Amanita is a toxic mushroom. The genus is responsible for approximately 95% of the fatalities resulting from **mushroom** poisoning. The most potent toxin present in **Amanita** mushroom is **α -amantin**.

Question Bank

1. Which of the following is not a Zoonotic disease?

- | | |
|------------------|------------|
| (a) Mucormycosis | (b) Rabies |
| (c) Plague | (d) SARS |

R.A.S./ R.T.S. (Pre) 2021

Ans. (a)

Zoonotic disease is an infectious disease caused by a pathogen (an infectious agent, such as a bacterium, virus, parasite or prion) that has jumped from an animal (usually a vertebrate) to a human. Rabies, Plague and SARS (Severe acute respiratory syndrome) are zoonotic diseases. Mucormycosis is not a zoonotic disease. Mucormycosis (previously called zygomycosis) is a serious but rare fungal infection caused by a group of molds called mucormycetes. These molds live throughout the environment. Mucormycosis mainly affects people who have health problems or take medicines that lower the body's ability to fight germs and sickness. It is also known as black fungus. Symptoms depend on where in the body the infection occurs. It most commonly infects the nose, sinuses, eye, and brain resulting in a runny nose, one-sided facial swelling and pain, headache, fever, blurred vision, bulging or displacement of the eye (proptosis), and tissue death. Other forms of disease may infect the lungs, stomach and intestines, and skin.

2. Ergotism is due to consumption of :

- | | |
|-------------------------|------------------------|
| (a) contaminated grains | (b) rotting vegetables |
| (c) contaminated water | (d) stale cooked food |

I.A.S. (Pre) 1998

Ans. (a)

Ergotism is a disease caused by consumption of contaminated grains. It is the effect of long-term ergot poisoning, traditionally due to the ingestion of the alkaloids produced by the Claviceps purpurea fungus that infects rye and other cereals.

3. 'Athlete's Foot' is a disease caused by :

- | | |
|---------------|--------------|
| (a) Bacteria | (b) Fungus |
| (c) Protozoan | (d) Nematode |

I.A.S. (Pre) 2001

Ans. (b)